Semantic News and Oil Price Database Project

This project turns oil-market, geopolitical risk, semantic news, and country exposure CSV files into a reproducible analytics workflow with MySQL tables, SQL views, and a transparent pricing model.

4,047
daily market rows

66,660
country-month GPR rows

1.516
test RMSE, USD

One workflow, three outputs

Source CSVs become a queryable database, analysis surfaces, and predictive model artifacts.

1
Datasets

Semantic news, prices, events, country impact

2
MySQL

Operational tables plus dim/fact reporting tables

3
Views

Daily oil/news features and event/country analysis

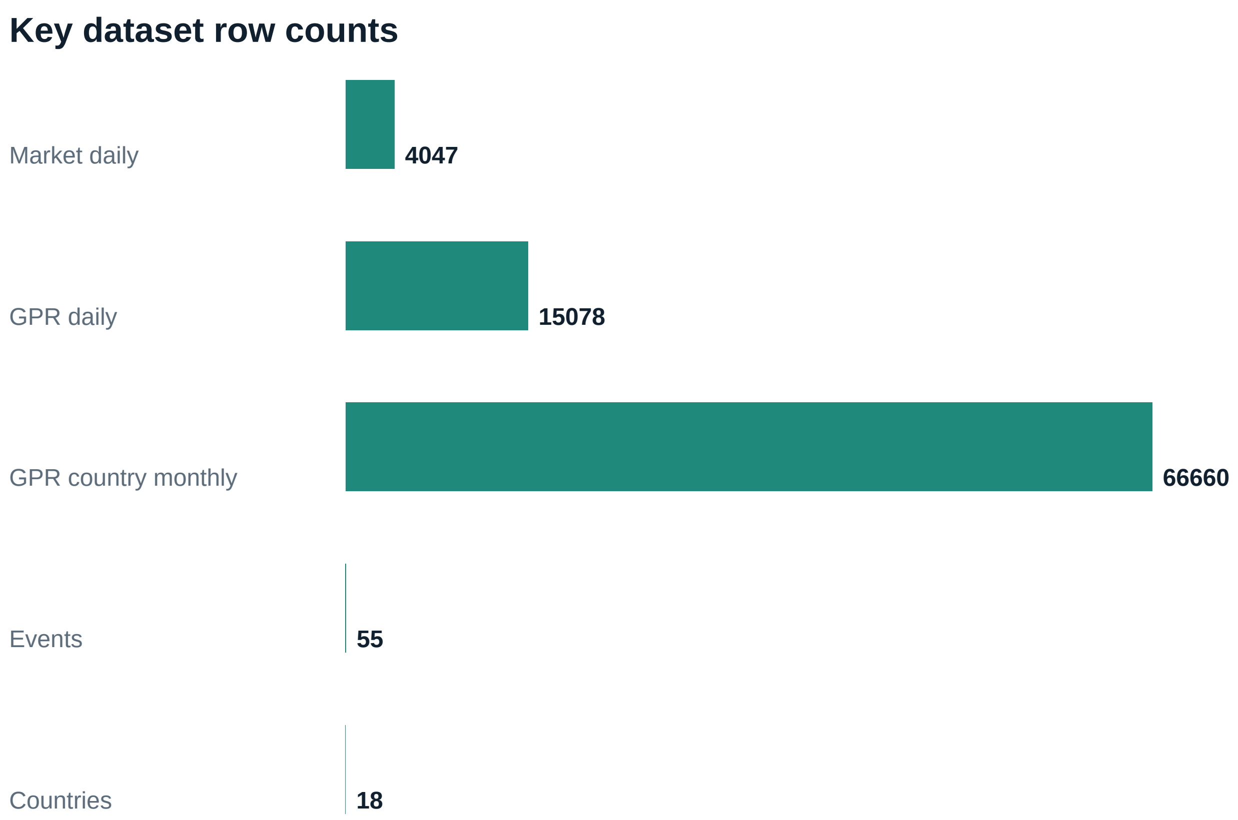
4
Model

StandardScaler -> Ridge

Final handoff includes scripts, schema docs, trained model, report, PDF, and deck.

The data mixes market prices with semantic risk

Coverage spans oil prices, GPR/news indices, events, and country-level exposure.



2010-2026

market table date span

1985-2026

daily GPR date span

The modeling table aligns Brent, WTI, volatility, GPR, event flags, and lag features for supervised learning.

Two database layers keep analysis flexible

Operational tables preserve source shape; dimensional tables support BI-style reporting.

Operational layer

- ops_market_daily
- ops_gpr_daily and monthly
- ops_events
- ops_countries and petrol snapshots

Dimensional layer

- dim_date, dim_country, dim_event
- fact_market_daily
- fact_gpr_daily and monthly
- fact_country_impact and petrol prices

Primary joins: market_date = gpr_date, market_date = event_date, country_id, and iso3.

Views turn raw tables into analysis surfaces

Three MySQL views package the joins needed for modeling, event study, and country impact analysis.

vw_daily_oil_news_features

Daily Brent/WTI, GPR, event, and volatility features for analysis and model inputs.

vw_event_price_reaction

Event-date oil price and risk indicators to inspect geopolitical market reactions.

vw_country_petrol_impact

Country exposure, petrol price snapshots, and vulnerability indicators in one reporting surface.

A transparent scikit-learn model predicts next-day Brent

Chronological split, standardized features, regularized linear model, exported metrics.

3,236
training rows

810
test rows

0.1
Ridge alpha

- Train through 2022-12-21; test through 2026-03-12
- Scale market, risk, volatility, lag, and event features
- Fit Ridge regression and save as a joblib pipeline
- Export JSON metadata and CSV test predictions

Features combine prices, risk, volatility, and events

The model uses 20 fields from ops_market_daily.csv.

2 current oil prices

2 macro indicators

1 GPR risk index

2 daily returns

6 lagged prices

4 volatility fields

1 Brent-WTI spread

2 event features

The model narrowly beats a strong baseline

Next-day oil pricing is hard because yesterday's price is already an excellent predictor.

Actual vs predicted Brent

Actual vs predicted test-set Brent prices

1.516
test RMSE, USD

1.118
test MAE, USD

0.967
test R-squared

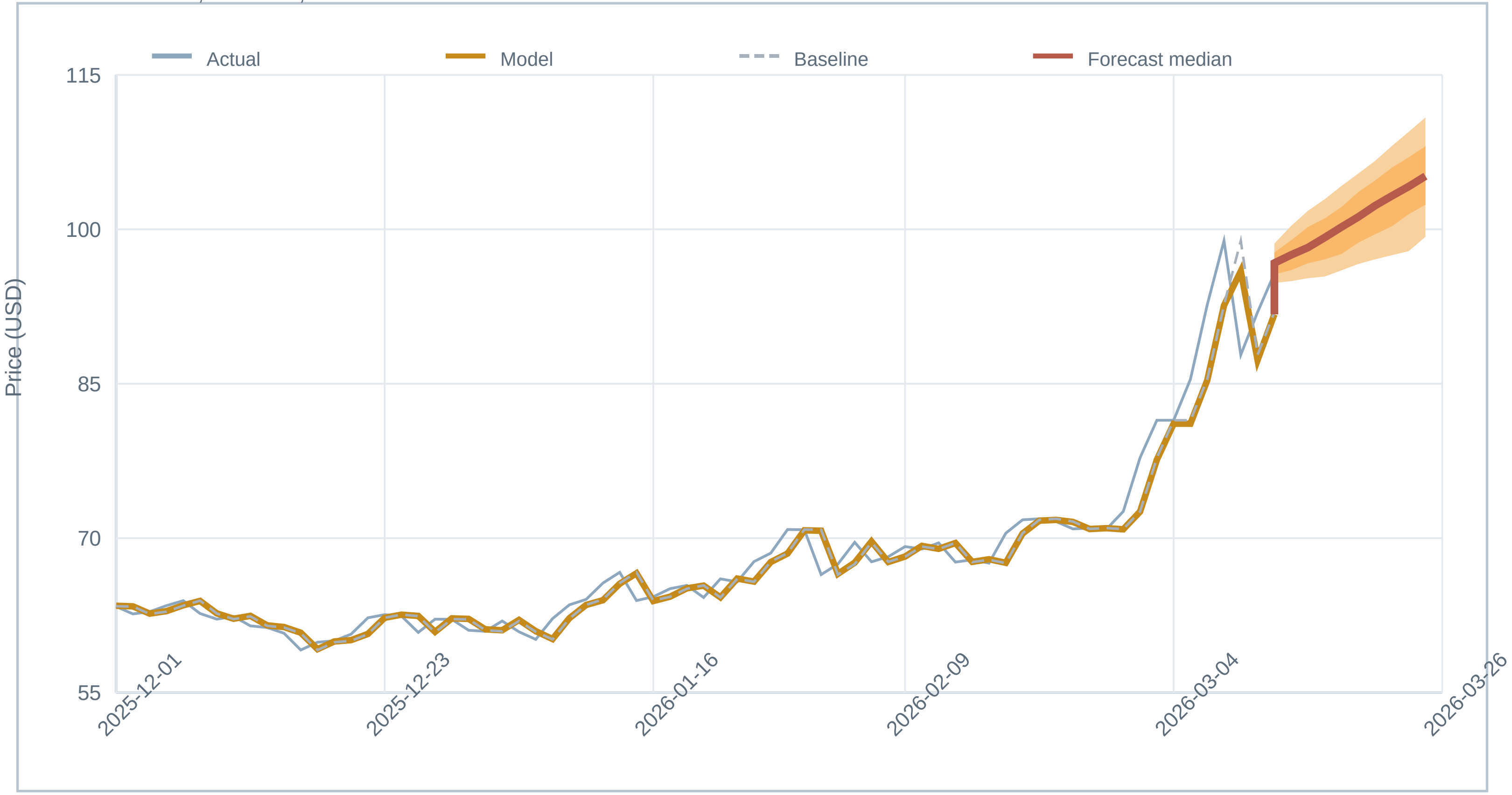
The diagonal shows perfect prediction. The cloud stays tightly clustered, which is what makes the result useful despite the small baseline gap.

The forecast extends from the test set

The median forecast and probability band bridge from the last observed trading day.

Test set plus 10-day forecast

Historical test set, baseline, and Monte Carlo forecast window



500
Monte Carlo paths

10
forecast trading days

P10-P90
outer forecast band

The shaded fan is the useful part for planning: it shows where the model is confident and where the path gets noisier.

The handoff is reproducible

Everything needed to load, train, predict, and present is in the workspace.

Run path

- python -m pip install -r requirements.txt
- python src\load_mysql.py --replace
- python src\train_oil_model.py
- python src\predict_oil_price.py

Deliverables

- MySQL loader and views
- Database schema guide
- Trained sklearn model
- Markdown/PDF report
- Editable deck source, PPTX, PDF, and HTML

Next improvements: foreign keys and indexes, dashboarding, rolling-window validation, and model prediction tables in MySQL.